

PALMDALE, CA, USA

WMO: 723820

Lat: 34.629N

Lon: 118.084W

Elev: 769

StdP: 92.42

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23182

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -3.7	(c) -2.2	(d) -18.4	(e) 0.8	(f) 8.9	(g) -16.3	(h) 1.0	(i) 8.6	(j) 14.4	(k) 9.0	(l) 12.6	(m) 9.4	(n) 1.6	(o) 200	(p) 0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.9	39.1	18.4	37.8	18.2	36.7	17.8	21.1	34.3	20.1	33.7	19.2	33.2	6.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.1	13.4	26.7	15.1	11.7	27.0	13.1	10.3	27.4	64.3	34.0	60.7	33.6	57.5	33.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.6	(b) 11.4	(c) 10.5	(e) -6.6	(f) 41.9	(g) 1.6	(h) 1.1	(i) -7.8	(j) 42.7	(k) -8.7	(l) 43.3	(m) -9.6	(n) 44.0	(o) -10.8	(p) 44.8		
(4) 12.6	(5) 11.4	(6) 10.5	(7) -6.6	(8) 41.9	(9) 1.6	(10) 1.1	(11) -7.8	(12) 42.7	(13) -8.7	(14) 43.3	(15) -9.6	(16) 44.0	(17) -10.8	(18) 44.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.2	8.0	9.2	12.4	15.1	19.6	24.5	28.4	27.8	24.0	17.7	11.6	7.3
	DBStd	8.16	3.22	3.32	3.61	4.12	4.24	3.88	2.68	2.57	3.65	3.57	3.77	3.29
	HDD10.0	274	77	51	18	6	0	0	0	0	0	1	26	96
	HDD18.3	1517	319	255	185	111	37	4	0	0	4	56	203	342
	CDD10.0	2899	16	29	93	160	298	437	569	553	420	238	73	12
	CDD18.3	1099	0	0	1	16	77	191	311	294	174	34	1	0
	CDH23.3	15468	2	9	86	319	1092	2624	4351	4035	2306	587	57	1
	CDH26.7	8580	0	1	13	97	501	1466	2621	2418	1256	203	5	0
(14) Wind	WSAvg	4.4	3.2	4.0	4.6	5.5	5.7	5.5	4.9	4.5	3.9	3.9	3.3	3.3
(15) Precipitation	PrecAvg	178	32	32	29	11	3	1	1	4	6	6	25	26
	PrecMax	395	120	163	133	63	42	18	13	37	54	52	143	134
	PrecMin	34	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	88	33	41	32	16	8	4	3	9	13	13	31	33
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	24.5	28.8	32.2	36.8	40.0	41.6	40.8	38.0	33.0	27.3	21.7
		MCWB	10.2	10.6	12.6	14.4	17.6	17.5	19.7	20.2	18.4	15.2	11.8	10.3
	2%	DB	19.4	22.0	25.9	29.3	34.0	37.6	39.6	38.8	36.5	31.1	25.0	18.7
		MCWB	8.8	9.6	11.9	13.1	16.0	17.4	18.7	18.6	17.9	14.5	10.9	9.0
	5%	DB	17.6	19.3	23.7	27.3	32.2	36.0	37.9	37.5	35.1	28.9	22.6	16.9
		MCWB	8.1	8.9	11.2	12.5	15.3	16.8	18.3	18.3	16.9	13.5	10.3	7.9
	10%	DB	15.4	17.0	21.3	24.5	29.8	34.0	36.6	36.2	33.4	27.0	20.0	14.3
		MCWB	7.4	8.1	10.5	11.6	14.1	16.3	17.9	17.6	16.5	12.7	9.4	7.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.1	12.6	14.2	15.7	19.1	20.9	22.5	22.5	21.8	17.8	13.7	12.8
		MCDB	17.4	17.9	24.3	28.3	32.4	34.3	35.6	35.6	32.9	25.6	22.3	17.8
	2%	WB	10.8	10.9	13.0	14.3	17.3	18.9	21.1	20.8	20.1	16.0	12.3	11.2
		MCDB	15.7	18.7	22.8	26.2	32.2	34.3	34.5	35.1	31.7	26.8	21.2	15.7
	5%	WB	9.8	10.0	12.0	13.2	15.8	17.9	20.2	19.8	19.0	14.8	11.2	9.7
		MCDB	14.5	17.3	21.4	24.7	29.9	33.3	33.9	34.4	30.8	25.6	19.8	14.0
	10%	WB	8.6	9.1	11.1	12.2	14.9	16.9	19.3	18.8	17.8	13.7	10.3	8.3
		MCDB	14.1	15.6	19.9	22.8	28.1	31.9	33.3	33.9	29.7	24.3	18.4	13.1
(35) Mean Daily Temperature Range	MDBR		14.0	13.3	14.4	14.8	15.1	16.3	15.9	16.5	16.8	16.0	15.4	13.5
	5% DB	MCDBR	17.6	18.2	18.6	18.8	18.9	18.5	17.6	17.8	18.2	19.1	19.0	17.1
		MCWBR	9.4	8.9	8.3	7.9	7.5	6.9	6.6	7.2	7.1	8.2	9.1	9.4
	5% WB	MCDBR	12.9	14.5	16.3	16.6	17.7	17.4	15.2	16.4	16.0	16.0	15.7	12.9
		MCWBR	7.5	7.5	7.8	7.4	7.3	6.8	6.6	7.4	6.7	6.9	7.9	7.8
(40) Clear-Sky Solar Irradiance	taub		0.286	0.296	0.314	0.335	0.346	0.345	0.365	0.347	0.334	0.321	0.288	0.284
	taud		2.521	2.510	2.447	2.392	2.388	2.401	2.393	2.420	2.453	2.468	2.543	2.548
	Ebn at Noon		932	958	962	950	937	932	912	925	927	914	921	913
	Edh at Noon		81	92	106	118	120	119	119	114	105	94	79	75
(44) All-Sky Solar Radiation	RadAvg		3.13	4.03	5.48	6.75	7.59	8.27	7.87	7.40	6.29	4.75	3.56	2.85
	RadStd		0.24	0.43	0.30	0.31	0.36	0.37	0.33	0.19	0.15	0.26	0.15	0.21

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+201	N/A
(47) Regional (46 neighbors)	+0.63	N/A	-1.78	+0.66	N/A	N/A	N/A	-115	+223	+106

Nomenclature: See separate page